Faculty of Engineering & Technology

Electrical & Computer Engineering Department

HARDWARE DESIGN LAB ENCS5131

Bug Report

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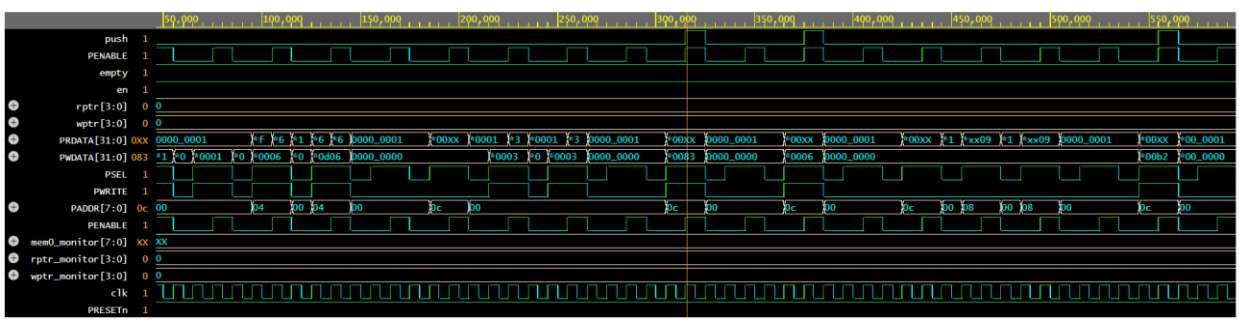
Section: 1

ID number	#1
Name	Data Written to FIFO Not Stored
Summary	Push operation fails to save to memory
Severity	CRITICAL
Priority	HIGH

Screenshot

```
UVM_INFO fifo_monitor.sv(69) @ 235000: uvm_test_top.env.agt.mon [fifo_monitor] MONITOR: addr=0x0 write=1 data=0x3
UVM_INFO fifo_monitor.sv(69) @ 265000: uvm_test_top.env.agt.mon [fifo_monitor] MONITOR: addr=0x0 write=1 data=0x3
UVM_INFO fifo_monitor.sv(69) @ 295000: uvm_test_top.env.agt.mon [fifo_monitor] MONITOR: addr=0x0 write=0 data=0x1
UVM_INFO fifo_monitor.sv(69) @ 325000: uvm_test_top.env.agt.mon [fifo_monitor] MONITOR: addr=0xc write=1 data=0x83
UVM_INFO fifo_monitor.sv(69) @ 355000: uvm_test_top.env.agt.mon [fifo_monitor] MONITOR: addr=0x0 write=0 data=0x1
UVM_INFO fifo_monitor.sv(69) @ 385000: uvm_test_top.env.agt.mon [fifo_monitor] MONITOR: addr=0xc write=1 data=0x6
UVM_INFO fifo_monitor.sv(69) @ 415000: uvm_test_top.env.agt.mon [fifo_monitor] MONITOR: addr=0x0 write=0 data=0x1
UVM_INFO fifo_monitor.sv(69) @ 445000: uvm_test_top.env.agt.mon [fifo_monitor] MONITOR: addr=0xc write=0 data=0x0
UVM_ERROR fifo_scoreboard.sv(256) @ 445000: uvm_test_top.env.sb [fifo_scoreboard] DATA MISMATCH (#1)! Expected=0x83 Got=0x0 count=1
UVM_INFO fifo_monitor.sv(69) @ 475000: uvm_test_top.env.agt.mon [fifo_monitor] MONITOR: addr=0x8 write=0 data=0x9
UVM_ERROR fifo_scoreboard.sv(156) @ 475000: uvm_test_top.env.sb [fifo_scoreboard] STATUS MISMATCH (#1)!
Expected: empty=0 full=0 almost=1 overflow=0 underflow=0 count=1
Actual:   empty=1 full=0 almost=1 overflow=0 underflow=0 count=0
UVM_INFO fifo_monitor.sv(69) @ 505000: uvm_test_top.env.agt.mon [fifo_monitor] MONITOR: addr=0x8 write=0 data=0x9
UVM_ERROR fifo_scoreboard.sv(156) @ 505000: uvm_test_top.env.sb [fifo_scoreboard] STATUS MISMATCH (#2)!
```

Waveform



Description

When writing data to the FIFO DATA register with FIFO enabled, the data does not get stored. Subsequent pop operations return incorrect values (typically 0x00), indicating the push operation failed to write data to memory.

PRESETn is HIGH during normal operation, but because the code incorrectly treats it as active high, the reset logic executes continuously, preventing any push or pop operations from working.

Steps to Reproduce

1. Write CTRL register: EN=1 (enable FIFO)
2. Write 0x83 to DATA register (address 0x0C) - Push operation
3. Read from DATA register (address 0x0C) - Pop operation
4. Observe returned value

Expected result

- Push 0x83 should store value in FIFO
- Pop should return 0x83

Actual result

- Push does not execute
- Pop returns 0x00 instead of 0x83

Issue in the code

The FIFO incorrectly treats PRESETn as active high, when it is actually active low. As a result, the reset branch executes on every clock edge during normal operation, resetting count, pointers, and empty/full flags and no push or pop logic actually run since the else branch is not entered. This prevents any data from being written to or read from the FIFO, effectively blocking all normal operations and making testing of push/pop functionality, overflow/underflow, and threshold flags impossible until the reset polarity is corrected.

```
// Count/flags
always @(posedge PCLK or posedge PRESETn) begin
  if (PRESETn) begin
    count <= '0;
    wptr  <= '0;
    rptr  <= '0;
    empty <= 1'b1;
    full  <= 1'b0;
  end else begin          // THIS WILL NOT EXECUTE
    if (clr) begin
      count <= '0;
      wptr  <= '0;
      rptr  <= '0;
      empty <= 1'b1;
      full  <= 1'b0;
    end else if (en) begin
      // Push
      if (push) begin
        if (!full) begin
          mem[wptr] <= PWDATA[7:0];
          wptr <= wptr + 1'b1;
          count <= count + 1'b1;
        end else begin
```

ID number	#2
Name	FIFO Count, Empty, and Full Registers Not Updating
Summary	count, empty, and full registers are continuously reset and do not reflect actual FIFO state
Severity	CRITICAL
Priority	HIGH

Screenshot

```
UVM_INFO fifo_monitor.sv(69) @ 475000: uvm_test_top.env.agt.mon [fifo_monitor] MONITOR: addr=0x8 write=0 data=0x9
UVM_ERROR fifo_scoreboard.sv(156) @ 475000: uvm_test_top.env.sb [fifo_scoreboard] STATUS MISMATCH (#1)!
Expected: empty=0 full=0 almost=1 overflow=0 underflow=0 count=1
Actual:   empty=1 full=0 almost=1 overflow=0 underflow=0 count=0
UVM_INFO fifo_monitor.sv(69) @ 505000: uvm_test_top.env.agt.mon [fifo_monitor] MONITOR: addr=0x8 write=0 data=0x9
UVM_ERROR fifo_scoreboard.sv(156) @ 505000: uvm_test_top.env.sb [fifo_scoreboard] STATUS MISMATCH (#2)!
Expected: empty=0 full=0 almost=1 overflow=0 underflow=0 count=1
Actual:   empty=1 full=0 almost=1 overflow=0 underflow=0 count=0
UVM_INFO fifo_monitor.sv(69) @ 535000: uvm_test_top.env.agt.mon [fifo_monitor] MONITOR: addr=0x0 write=0 data=0x1
UVM_INFO fifo_monitor.sv(69) @ 565000: uvm_test_top.env.agt.mon [fifo_monitor] MONITOR: addr=0xc write=1 data=0xb2
```

Description

The FIFO internal counter (count) and status flags empty and full do not update as expected:

count remains at 0, empty remains high and full remains low.

These registers are reset in the main FIFO always block, which incorrectly treats PRESETn as active high and since PRESETn is HIGH during normal operation, the reset branch executes on every clock cycle, preventing any updates to count, wptr, rptr, empty, and full.

Steps to Reproduce:

1. Write CTRL register: EN=1 (enable FIFO)
2. Observe count, empty, and full registers
3. Write values to DATA register (address 0x0C)
4. Observe count, empty, and full again

Expected Result:

- count increments with each push
- empty goes low when FIFO has data
- full goes high when FIFO reaches capacity

Actual Result:

- count remains 0
- empty stays high
- full stays low

Additional Issues Identified by Code Inspection

Issue A: Incorrect FIFO Full Condition

Description:
The FIFO full flag is asserted when count == DEPTH-1:

full <= (count == DEPTH-1);

For a FIFO of depth DEPTH, the FIFO should be considered full when count == DEPTH.
The current implementation flags the FIFO as full while one storage location is still available.

Issue B: DROP_ON_FULL Control Has No Functional Effect

Description:
The DROP_ON_FULL control bit does not change behavior when the FIFO is full:

```
    if (drop_on_full) begin
      // drop silently but set flag
      overflow <= 1'b1;
    end else begin
          overflow <= 1'b1;
    end
  end
end
```

Both branches perform the same action. As written, enabling or disabling DROP_ON_FULL has no functional impact on overflow handling.

Issue C: Overflow and Underflow Flags Are Not Cleared

Description:
The overflow and underflow flags are not included in the clear logic.

- They are not reset by PRESETn
- They are not cleared by CTRL.CLR

```
if (clr) begin
  count <= '0;
  wptr  <= '0;
  rptr  <= '0;
  empty <= 1'b1;
  full  <= 1'b0;
end else if (en) begin
```

Why These Issues Are Currently Blocked

Due to the incorrect reset polarity, the FIFO remains effectively in a continuous reset state during normal operation. As a result, core state such as count, read/write pointers, and empty/full flags are reinitialized every clock cycle, preventing any push or pop operations from executing. Because overflow, underflow, and enable-related behaviors depend on successful push/pop activity, these conditions can never be triggered or observed in simulation. Resulting in the main features being functionally blocked and preventing validation or reproduction until the reset polarity issue is corrected.